

Mapping of Course Outcomes and Topics to the Term Project

CISCaRL — Conformalized Interpretable Survival Causal Rule Lists

Course: 23CSE301 — Machine Learning (Semester V, AY 2026–27)

Programme: B.Sc. CSE, Amrita School of Computing, Bengaluru

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1. Project at a Glance

CISCaRL estimates **heterogeneous treatment effects** (who benefits from a treatment and by how much) from **right-censored survival data**, and returns a short, human-readable **rule list** where every rule carries a treatment-effect estimate, a finite-sample-valid 90% conformal interval, a bootstrap stability score, and a treat/avoid recommendation. It addresses a real gap: causal survival forests are accurate but opaque (black boxes), while rule-based alternatives emit hundreds of rules and no uncertainty. The project is implemented end-to-end in Python (scikit-learn, lifelines, scikit-survival), benchmarked fairly against nine methods, validated on synthetic data with known ground truth, and demonstrated on the real ACTG175 randomized HIV trial.

2. Mapping to Course Outcomes (CO1–CO5)

CO	Course Outcome	How the project addresses it
CO1	Understand the fundamental concepts, issues and challenges of machine learning	The project is built around a foundational ML problem: estimating the conditional average treatment effect (CATE) under confounding and censoring. It explicitly studies the classical issues — generalization (underfit/regular/overfit, measured via the train–test gap), explainability (the black-box CSF problem is the motivation), algorithmic bias (subgroup disparity analysis), and the curse of dimensionality (a $p=10$ vs $p=50$ experiment).
CO2	Implement machine learning algorithms using programming tools and provide a solution to real-world applications	A complete, reproducible Python implementation (<code>cis_carl.py</code> , <code>scre.py</code> , <code>experiments.py</code> , <code>fair_benchmark.py</code> , <code>utils.py</code> , <code>data.py</code>) using scikit-learn, lifelines and scikit-survival. It is applied to a real-world medical problem and demonstrated on the real ACTG175 randomized controlled trial. A one-command reproducible run is provided (<code>run_all.py</code>) with a full reproduction manual.
CO3	Apply machine learning algorithms for a parameter-estimation or prediction problem	Uses a doubly-robust (DR) learner to estimate the per-patient treatment-effect parameter (pseudo-ITE) and predicts the CATE at a chosen horizon t^* . Prediction quality is quantified with regression metrics RMSE, MAPE and R^2 , reported for every method in the benchmark tables.
CO4	Apply supervised learning techniques for a classification problem and analyse their performance	Supervised learners (Random Survival Forest, Gradient Boosting, Cox) are trained and analysed. The treatment decision is recast as classification (treat if predicted CATE > 0) and evaluated with precision, recall, accuracy, F1 and AUC. Performance is analysed with paired significance tests (Holm-corrected) and effect sizes across 48 settings.
CO5	Apply unsupervised and ensemble learning techniques to solve a given problem	The method is fundamentally ensemble -based: Random Forest (bagging) and Gradient Boosting (boosting) generate the candidate rules, and bootstrap-aggregated stability

		selection is itself an ensemble-style aggregation. The rule list provides an interpretable unsupervised partitioning of patients into subgroups.
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3. Mapping to Syllabus Topics

Legend: **Y** = fully covered, **P** = partially, **–** = not used (stated honestly).

Unit	Topic	?	Where it appears in the project
1	Types of ML, features, class boundary	Y	Supervised CATE regression; rule conditions are decision boundaries
1	Training/Validation/Testing; Generalization (underfit / regular / overfit)	Y	70/30 train-test split plus a 70/30 rule-train/calibration split; generalization gap measured (0.002–0.008 → regular fit)
1	Loss / Cost function	Y	Lasso penalised mean-squared error in rule selection; Cox partial likelihood for the competing method
1	Explainability	Y	The central contribution — an interpretable rule list replaces the black box
1	Algorithmic bias	Y	Subgroup-disparity analysis of error across covariate-defined groups
1	Data and algorithmic privacy	P	Handling of medical data discussed; no dedicated privacy mechanism
1	Curse of dimensionality	Y	Robustness experiment at p=10 vs p=50 (40 noise features)
1	Dimensionality reduction — PCA, LDA	–	Not used (conformal rule selection operates in the original space); noted as future work
1	Feature selection — sequential & bi-directional	P	Rule/stability selection and univariate rule screening are selection procedures, though not classical SFS/SBS
1	k-Nearest Neighbour classifier	–	Not used by CISCaRL (covered separately in the course lab)
1	Regression: linear, logistic; Regularization LASSO / Ridge	Y (LASSO)	LASSO is used directly for sparse rule selection; propensity estimated with a classifier; Ridge is not central
1	Classifier metrics — precision, recall, accuracy, F-score, AUC	Y	Computed for the treat/avoid decision and compared across methods
1	Regression metrics — RMSE, MAPE, R ²	Y	Computed for every method; R ² is discussed and PEHE (RMSE) used as the primary CATE metric
2	Decision Trees	Y	Candidate rules are extracted as IF/THEN paths from tree models
2	Support Vector Machines	–	Not used
2	Naive Bayes	–	Not used
2	Markov model / Hidden Markov Model	–	Not used
2	Artificial Neural Networks (perceptron, MLP, back-propagation)	–	Not used

2	Parameter estimation — MLE, Bayesian, EM	P	Maximum-likelihood style estimation via the Cox partial likelihood and the doubly-robust estimator; no Bayesian/EM component
3	Clustering — hierarchical, K-means, Elbow technique	–	Not used; the rule list provides an interpretable partition instead of a clustering
3	Ensemble learning — bagging, boosting, AdaBoost, Random Forest	Y	Core of the method: Random Forest + Gradient Boosting generate rules; bootstrap aggregation performs stability selection
3	Introduction to Reinforcement Learning	–	Not used
3	Hyper-parameter tuning; cross-validation	Y	LassoCV (cross-validated LASSO); tuned stability threshold, bootstrap count, shrinkage and minimum support

4. Mapping to the Term-Project Evaluation Rubric

Evaluation component	CO	Evidence in the project
Viva — Data Modelling	CO1	Dataset provenance (PBC, SUPPORT, GBSG, ACTG175), semi-synthetic design with known CATE, preprocessing and DR-learner outcome construction
Viva — Model Analysis	CO3, CO5	Metric tables, significance tests, coverage validation, hyper-parameter sensitivity
Project — Data Modelling	CO1	Reproducible data loaders and pipeline (<code>data.py</code> , <code>utils.py</code> , <code>REPRODUCE.md</code>)
Project — Code Implementation	CO2	Modular Python package plus <code>run_all.py</code> single-command demonstration
Project — Results Analysis	CO4	Fair benchmark across 9 methods $\times$ 4 datasets $\times$ 4 DGPs, both effect regimes
Project — Inferences	CO4	Conclusions on the accuracy–interpretability trade-off and honest limitations

5. Topics Not Covered (Honest Scope Statement)

The project deliberately does not use SVM, Naive Bayes, hidden Markov models, artificial neural networks, dedicated clustering (K-means / hierarchical / Elbow), reinforcement learning, or explicit dimensionality reduction (PCA / LDA). These are not part of the HTE survival problem the project targets; where a classical technique is relevant (regularization, ensembles, decision trees, cross-validation, hyper-parameter tuning), it is used directly and is central to the method.

6. Summary

CISCaRL maps cleanly onto the course: it exercises foundational ML concepts and challenges (CO1), delivers a full real-world implementation (CO2), performs parameter estimation and prediction (CO3), applies and analyses supervised classification-style decisions (CO4), and is built on ensemble learning (CO5). It demonstrates the syllabus topics that matter for the problem — generalization, loss/cost functions, explainability, bias, curse of dimensionality, LASSO regularization, classification and regression metrics, decision trees, ensembles and hyper-parameter tuning — while honestly stating the topics it does not use.